

Series Operation and Parallel Operation of SCR

SERIES & PARALLEL OPERATION OF THYRISTORS

SCRs are connected in series in order to meet the hv demand & in parallel for fulfilling the high current demand.

STRING EFFICIENCY

It is a parameter for measuring the degree of utilization of SCRs in a string.

$$\text{String efficiency} = \frac{\text{Actual voltage/current rating of whole string}}{\text{Individual voltage/current rating of each SCR} \times \text{No. of SCR in string}}$$

In practice, this ratio is less than 1. For obtaining the highest possible string n , the SCRs connected in the string must have identical I-V characteristics.

DERATING FACTOR

It is a measure of the reliability of a string.

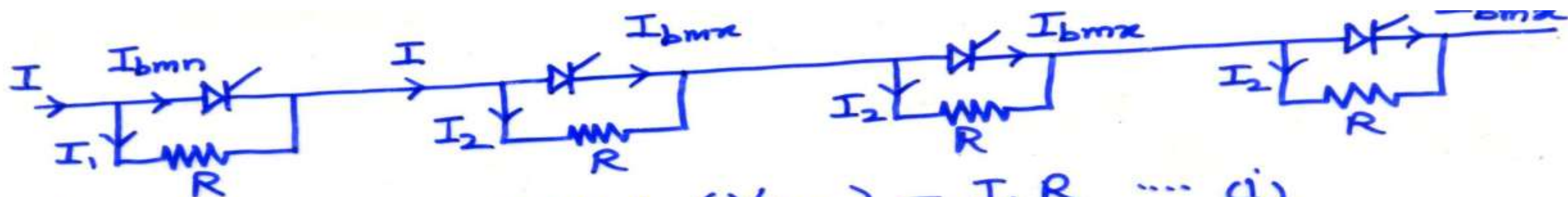
$$\text{DRF} = 1 - \text{string } n$$

Greater the magnitude of DRF, greater is the reliability.

SERIES OPERATION

On account of inherent variations in their characteristics, voltage shared by each SCR in a series circuit may not be equal. A uniform voltage distribution in steady state can be achieved by connecting a suitable resistance across each SCR such that each parallel combination has the same resistance.

Practically it is more feasible to connect the same value of shunt resistance across each SCR. Such a resistance is called static equalizer & the arrangement is termed static equalizing circuit.



Voltage across SCR 1 (V_{bm}) = $I_1 R$ (i)

Voltage across $(n-1)$ SCRs = $(n-1)I_2 R$

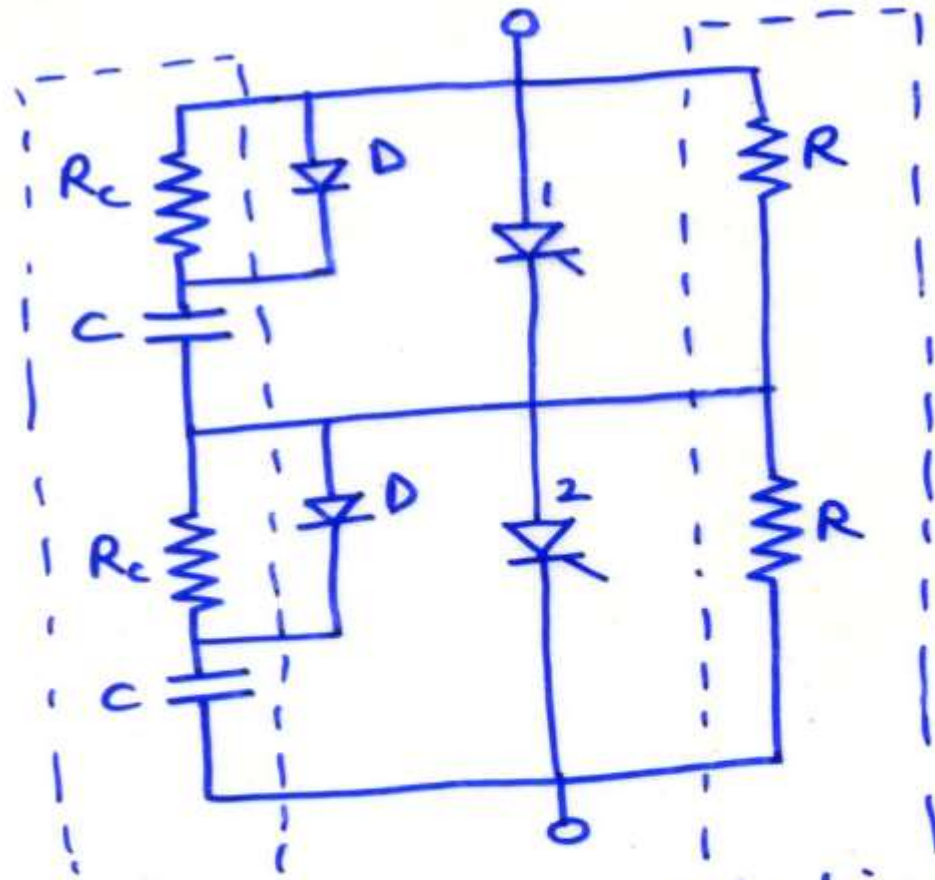
$$\begin{aligned}
 V_s &= I_1 R + (n-1) R I_2 \\
 &= V_{bm} + (n-1) R (I - I_{bmx}) \\
 &= V_{bm} + (n-1) R [I_1 - (I_{bmx} - I_{bmn})] \\
 &= V_{bm} + (n-1) R [I_1 - \Delta I_b] \\
 &= V_{bm} + (n-1) R I_1 - (n-1) R \Delta I_b \dots (ii)
 \end{aligned}$$

From eq. (i) & (ii)

$$R = \frac{nV_{bm} - V_s}{(n-1) \Delta I_b}$$

The voltage distribution during turn on & turn off would be unequal due to self capacitances. A resistor can't maintain equal voltage distribution under transient condition. Voltage distribution, under such conditions, is obtained by employing shunt capacitors.

The resultant of shunt capacitance & self capacitance of each SCR tends to be equal.



Dynamic
Equalizing
Circuit

Static
Equalizing
Circuit

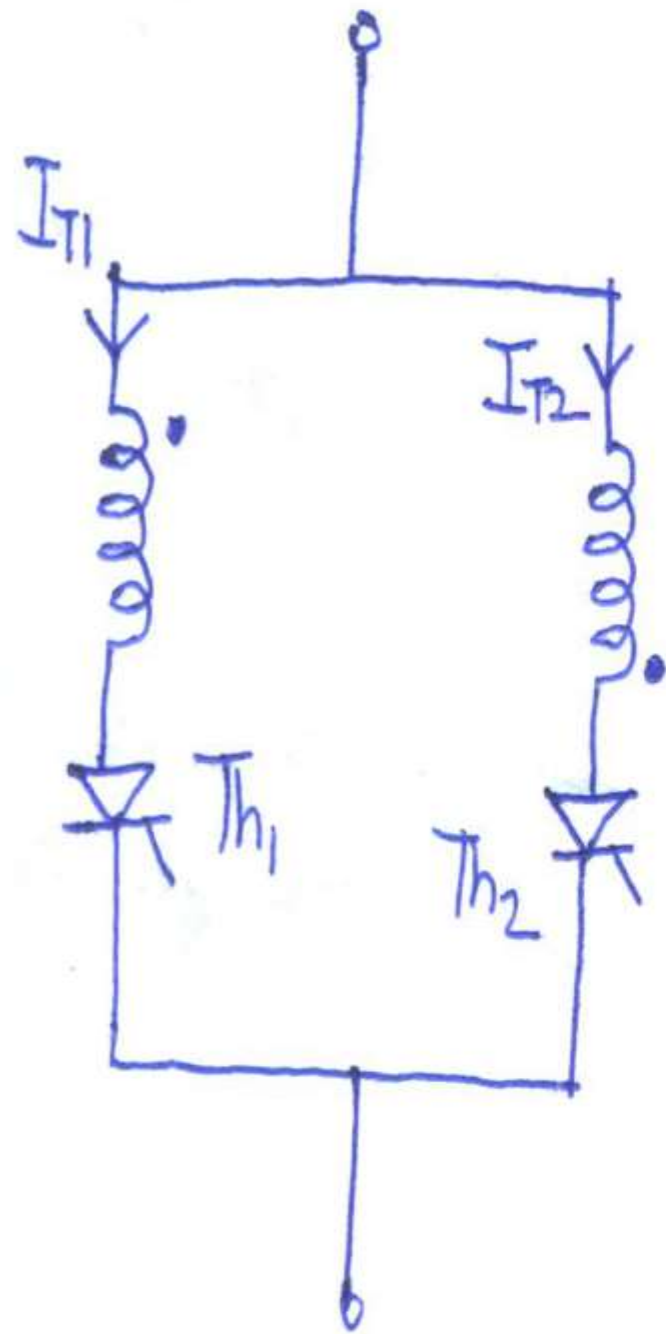
$$C = \frac{(n-1) \Delta Q}{nV_{bm} - V_s}$$

where $\Delta Q \rightarrow$ excess charge left
in SCR
 $n \rightarrow$ number of SCR
 $V_s \rightarrow$ string voltage
 $V_{bm} \rightarrow$ Highest transient
Voltage

PARALLEL OPERATION

The dynamic resistances of thyristors in parallel may not be equal & this would cause unequal currents through them. If current through one of them is more, its internal power dissipation will be more. This will raise its temp, & lower its dynamic resistance & increase the current through it. This cumulative process is called thermal runaway.

In dc circuits, effect of unequal currents can be minimized by connecting resistances in series with the thyristors. The resistance values are so selected that total anode to cathode voltage across each branch is same. However, the use of resistances, entail considerable power loss.



In case of unequal thyristor currents, unequal voltages are induced in the inductors, which reduce the difference between I_{T1} & I_{T2} .